**Introduction to Software Engineering**

# Software Requirements Specification

The student team is required to complete the **Software Requirements Specification** documentation for the assigned course project, following the attached template.




Software Engineering Department
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# Software Requirements Specification

## Objectives

This document focus on the following topics:

- ✔ Completing the Software Requirements Specification (SRS) document with the following sections:
  - Problem Statement
  - Stakeholders
  - Functional, non-functional and domain requirements
  - Use case model and use case specifications
  - Prototype / mockups
  - Traceability and requirements review
- ✔ Including an AI usage log for this phase
- ✔ Understanding the Software Requirements Specification (SRS) document.

# 1 Member Contribution Assessment

PA2 contribution allocation confirmed on 02 Nov 2026: 25% for each member, totaling 100%. Evidence links identify assigned work; completion and review status are recorded separately.

| ID | Name | Role(s) | Contribution (%) | Evidence |
|---|---|---|---|---|
| 24127169 | Trương Lê Trung Hiếu | PM / technical coordination | 25% | DID-21, DID-43, DID-45, DID-56-58 |
| 24127130 | Trần Minh Tiến | Business Analyst | 25% | DID-40-42, DID-44, DID-46, DID-49, DID-55 |
| 24127020 | Nguyễn Thành Đạt | Designer / UX | 25% | DID-50-52, Figma |
| 24127035 | Võ Đình Cao Minh Hào | Tester / requirements QA | 25% | DID-47-48, DID-53-54 |
| Total | | | 100% | |

# 2 Introduction

## 2.1 Purpose and Scope

*DiDung is a responsive web service for accessing air-quality information and short-term PM2.5 forecasts for supported locations in Ho Chi Minh City. The MVP includes data ingestion, an AQI/PM2.5 dashboard, forecast views, account preferences, configurable thresholds, and in-app notifications. Flood and high-tide prediction are feasibility-study topics only, not committed MVP functions. Target readers: users, BA, designer, developer and tester; course staff are external readers only. PA1 is the latest scope baseline where it supersedes PA0. TA is not a project participant or approver. Current BA feedback on open requirements and decisions remains pending.*

## 2.2 Glossary

| Term | Definition |
|---|---|
| AQI | Air Quality Index: a numerical index derived from pollutant data under a specified standard. |
| PM2.5 | Fine particulate matter with aerodynamic diameter at most 2.5 micrometres. |
| MVP | Minimum Viable Product: minimum approved release scope. |
| FR / NFR | Functional / Non-functional Requirement. |
| MoSCoW | Must, Should, Could, Won't for this release. |
| Freshness | Time elapsed since the observation or model output timestamp. |
| Forecast | Estimated future pollutant values with stated horizon and provenance. |
| SRS | Software Requirements Specification. |
| UC / AC | Use Case / Acceptance Criteria. |
| RTM | Requirements Traceability Matrix. |
| Stale data | Data older than the team-approved freshness threshold (threshold pending). |

# 3 Problem Statement

*Residents and commuters in Ho Chi Minh City need accessible, understandable information about current air quality and near-term exposure risks. Existing environmental feeds vary in geographic coverage, observation versus modeled provenance, refresh intervals and completeness. DiDung aims to aggregate available PM2.5 and relevant weather information, communicate AQI with source/timestamp, provide short-term PM2.5 forecasts and support user-selected locations and alert preferences. The intended system is a responsive Vietnamese web application using the proposed Next.js UI/API, Supabase PostgreSQL/Auth and Vercel hosting. Constraints include public API quotas, incomplete or stale observations, model uncertainty, limited course time and privacy/security. It must not imply street-level measurement or medical certainty. Facebook-derived flood/tide risk is a separate feasibility study and will not be advertised as an MVP forecast without a documented scope change and verification.*

# 4 Stakeholders

*Stakeholders are proposed from PA0/PA1 and team analysis; no interview is claimed unless evidence is attached. The PA2 scope is AQI/PM2.5-first and follows PA1 where it supersedes PA0. TA is not a project participant or approver. BA feedback on open requirements and decisions remains pending.*

| # | Stakeholder | Description | Interests / concerns |
|---|---|---|---|
| 1 | Course staff (external reader) | May read submitted coursework; does not participate in project decisions or review. | Clarity and traceability in submitted work; no project approval role. |
| 2 | Resident / visitor | Views air quality in supported locations. | Current readings, reliable source/time labels and clear missing-data states. |
| 3 | Commuter | Checks selected places and near-term forecasts. | Accessible comparisons and clearly labeled uncertainty. |
| 4 | Registered user | Sets tracked locations and alert thresholds. | Account security, correct preferences and in-app notices. |
| 5 | External data provider | Supplies permitted pollutant/weather data. | Proper attribution, quota compliance and responsible usage. |
| 6 | DiDung project team | BA, PM, designer, developer and tester. | Feasible requirements, review evidence, implementable design and testability. |

# 5 Requirements

## 5.1 Functional Requirements

*Functional requirements below are draft normative statements. Must priorities follow PA1 Scope A; BA feedback on open requirements remains pending. TA review or approval is not part of this project. Listed sources are documents or team analysis, not unrecorded interviews.*

| ID | Requirement | Feature | Priority | Source |
|---|---|---|---|---|
| FR-01 | *The system shall retrieve available PM2.5 and weather inputs from configured permitted sources, recording source, location, unit and timestamp.* | F01 | Must | PA1 F01; data source study DID-43 |
| FR-02 | The system shall store validated observations or modeled inputs and flag missing, invalid or stale records rather than silently substituting values. | F01 | Must | PA1 F01; BA assumption pending |
| FR-03 | The system shall display available AQI/PM2.5 for a selected supported location with source, timestamp and data status. | F07 | Must | PA1 F07; residents |
| FR-04 | The system shall display location-related AQI/PM2.5 on a map and provide available historical values for the selected location/time range. | F07 | Must | PA1 F07; commuters |
| FR-05 | The system shall present a short-term PM2.5 forecast with forecast horizon, issue time and distinction from observed values. | F02 | Must | PA1 F02; residents |
| FR-06 | The system shall register and authenticate users using a configured identity service and prevent unauthorized access to account settings. | F05 | Must | PA1 F05; security requirement |
| FR-07 | The system shall let authenticated users save, edit and delete tracked locations and personal alert thresholds. | F05 | Must | PA1 F05; registered users |
| FR-08 | The system shall evaluate eligible air-quality data against a user's active alert threshold and create an in-app notice when configured trigger conditions occur. | F06 | Must | PA1 F06; registered users |

| ID | Requirement | Feature | Priority | Source |
|---|---|---|---|---|
| FR-09 | The system shall display a user's notifications and allow them to mark individual notices as read. | F06 | Must | PA1 F06; registered users |
| FR-10 | The system shall present an explicit error, empty or stale state when requested air-quality data are unavailable or outdated. | F07 | Must | PA1 F07; error-handling analysis |
| FR-11 | The system should present available baseline-versus-ML forecast evaluation results with clearly identified metrics and datasets. | F03 | Should | PA1 F03 |
| FR-12 | The system could display general, non-medical interpretation guidance corresponding to the selected AQI standard. | F08 | Could | PA1 F08 |
| FR-13 | The system should provide system administrators with a monitoring view of pipeline run status, stale-data warnings, deployment health and forecast-error summaries. | F09 | Should | PA1 F09 (Should; latest scope; PA0 nice-to-have is historical) |

## 5.2 Non-Functional Requirements

*All numerical NFR thresholds below are PROPOSED test targets, not measured results or stakeholder-approved SLAs. Tester/PM must check feasibility and define the test environment before baselining.*

| ID | Category | Requirement (measurable) | How it will be verified |
|---|---|---|---|
| NFR-01 | Performance | Proposed target: under 50 simulated concurrent users, cached dashboard GET requests shall complete in <=2s at P95, excluding third-party data refresh. | Run repeatable load test; report environment, payload, P95 and failure rate. |
| NFR-02 | Security | Protected account/preferences APIs shall return 401/403 for unauthenticated/unauthorized requests in 100% of enumerated security test cases. | Integration/auth tests on protected routes; review RLS policies. |
| NFR-03 | Usability | At least 4 of 5 representative users shall complete location selection and AQI | Usability task test with recorded times, |

| ID | Category | Requirement (measurable) | How it will be verified |
|---|---|---|---|
| | | inspection unaided within 60 seconds in a moderated test (proposed target). | environment and consent. |
| NFR-04 | Accessibility | Primary text and control states shall meet WCAG 2.2 AA contrast targets: 4.5:1 normal text, 3:1 large text and required non-text indicators where applicable. | Contrast audit of design and implemented screens; keyboard testing on implementation. |
| NFR-05 | Reliability | When a provider returns an error, the UI shall display a distinct unavailable/stale state, never a newly fabricated measurement, in 100% of scripted failure tests. | Inject API 4xx/5xx/timeouts and inspect screen/API output. |
| NFR-06 | Data integrity | All displayed pollutant values shall retain source and timestamp metadata in 100% of synthetic ingestion tests. | Schema/check ingestion tests for null/units/timezone/provenance. |
| NFR-07 | Responsive design | For viewport widths 390px and 1440px, main user flows shall show no horizontal scrolling caused by layout overflow in the defined screen set. | Browser responsive audit; document browsers and exceptions. |
| NFR-08 | Privacy | User location preferences and notification records shall not be readable by another account in any of the account-isolation test cases. | Two-user RLS/API isolation test, no credentials/tokens in evidence. |

## 5.3 Domain Requirements and Business Rules

| ID | Rule | Source (policy, law, standard) |
|---|---|---|
| BR-01 | The team shall declare one chosen AQI scale and calculation/interpretation source; it must not mix scale categories without labeling. | Approved AQI standard/policy - pending selection |
| BR-02 | Observations and forecasts must be labeled distinctly; every datum must expose source and observation/issue time. | PA1 data provenance; team decision pending |
| BR-03 | Staleness and forecast horizon thresholds must be defined per source/feature before acceptance testing. | Provider documentation; proposed team rule |

| ID | Rule | Source (policy, law, standard) |
|---|---|---|
| BR-04 | Missing measurements must never be silently replaced with zero or a current reading from an unrelated location. | Domain data quality rule |
| BR-05 | Threshold-based notifications shall use the configured comparator, avoid unintended duplicates and record trigger time. | Stakeholder needs; team rule pending |
| BR-06 | Only the account owner may access or change saved preferences and in-app notification history. | Account security/privacy policy |
| BR-07 | Flood/tide forecasts from social posts remain outside MVP until feasibility and formal change-control approval. | PA1 Scope A; PA1 is the latest baseline where it supersedes PA0. |

## 5.4 Acceptance Criteria

*Draft acceptance criteria (Given / When / Then):*

*AC-F01-01: Given a configured, permitted source returns a well-formed pollutant or weather record with its source, location, unit and timestamp, when ingestion validates and stores the record, then the stored record retains those fields and is marked valid; a modeled value is not relabeled as observed.*

*AC-F01-02: Given the source input is absent or fails a required field/type/unit validation, when ingestion processes the input, then it is marked rejected/invalid, no new valid measurement is created from it, and the rejection reason is recorded.*

*AC-F01-03: Given one or more configured permitted sources return available PM2.5/weather records for a supported location, when the scheduled ingestion run executes, then it retrieves the available inputs and records the run outcome and source/location for each retrieved record. Exact provider coverage and update cadence remain subject to DID-43/source validation.*

*AC-F01-04: Given an input timestamp is older than the approved freshness threshold for that source, when ingestion validation evaluates the input, then the record is flagged stale and is not silently relabeled as current or replaced by a value from another location. The approved source-specific threshold remains unspecified (BR-03).*

*AC-F01-05: Given a configured source API returns HTTP 4xx/5xx or times out, when a scheduled ingestion attempt calls that source, then the attempt is marked failed with source and attempt time, no new valid measurement is created from that failed response, and previously stored valid data are not overwritten by fabricated data. Retry/backoff limits remain implementation design for PA3/PA4.*

*AC-F02-01: Given a supported location has an available PM2.5 forecast whose target horizon is from 1 through 6 hours, when a Web User opens the forecast, then the estimate, forecast horizon, issue time, source/provenance and forecast label are shown distinctly from observations.*

*AC-F02-02: Given no forecast output is available, including because the configured forecast/provider API is unavailable, returns an error or times out, when a Web User opens or refreshes the forecast, then the UI shows an explicit unavailable state and does not create or display a fabricated forecast curve/value.*

*AC-F05-01: Given the configured identity service recognizes valid credentials, when the Web User signs in, then authentication succeeds and the protected preferences view is available only to that authenticated account.*

*AC-F05-02: Given credentials are invalid or rejected by the configured identity service, when the Web User submits sign-in, then access is denied with a non-secret error response and no protected preferences are disclosed.*

*AC-F05-03: Given an authenticated user selects a supported location and provides a valid threshold under the currently configured rule, when the user saves the preference, then it persists under that user's account and is returned as that user's saved preference. The AQI scale/comparator must be configured before acceptance (BR-01/BR-05).*

*AC-F05-04: Given a user submits a registration request with details accepted by the configured identity service, when registration completes, then an account is created by that service and can authenticate through the same service. Provider-specific validation/verification remains as defined by the selected provider.*

*AC-F05-05: Given a requester is unauthenticated, or an authenticated requester does not own the target account/preferences, when the requester reads or changes protected settings, saved locations or alert thresholds, then the request returns an unauthorized/forbidden result (401/403 as applicable), no protected data are disclosed, and the owner's stored preference remains unchanged.*

*AC-F05-06: Given an authenticated user owns an existing saved location or alert threshold, when the user edits or deletes that preference, then the persisted record reflects the edit or is absent after deletion, and unrelated account preferences remain unchanged.*

*AC-F06-01: Given an active user threshold has a saved comparator, a supported location has an eligible fresh value, and the value crosses the saved threshold rule, when the threshold evaluator processes that event, then exactly one in-app notice is recorded for the event with its location and trigger time. The comparator and equality boundary are not chosen here; they remain a BR-05 decision.*

*AC-F06-02: Given an event has already produced an in-app notice and the same event identity is submitted again, when duplicate suppression processes the replay, then the total number of notices for that event remains exactly one. The event identity and suppression window remain BR-05 decisions.*

*AC-F06-03: Given an authenticated user owns an unread notification, when the user marks it read, then its persisted read status is `read` and remains `read` after the notification center is reopened.*

*AC-F06-04: Given an authenticated user owns one or more in-app notifications, when that user opens the notification center, then only that user's notifications are displayed with their stored read/unread state.*

*AC-F06-05: Given a fresh eligible value does not satisfy the user's saved comparator/threshold rule, when the evaluator processes the value, then zero notifications are created for that evaluation. The equal-to-threshold case follows the BR-05 boundary decision and remains pending until that decision is recorded.*

*AC-F06-06: Given an input is flagged stale or invalid under the configured validation rules, when the threshold evaluator processes it, then zero notifications are created from that input and the data quality status remains available for inspection. The eligibility rule requires BA/Tester confirmation before acceptance sign-off.*

*AC-F06-07: Given authenticated user B does not own a notification belonging to user A, when B attempts to read or change A's notification/read status, then the request returns 401/403 as applicable, exposes no notification content, and leaves A's notification unchanged. This aligns with NFR-02/NFR-08 and BR-06.*

*AC-F07-01: Given a supported location has a valid fresh AQI/PM2.5 value, when a Web User selects that location, then the value, source, timestamp and observed/modeled data status are shown.*

*AC-F07-02: Given a requested value is older than the approved freshness threshold, when a Web User selects the location, then the UI labels the value stale, displays its true timestamp and does not present it as current. The freshness threshold remains BR-03/TBD.*

*AC-F07-03: Given a selected location is unsupported or has no valid data, when the Web User requests its AQI/PM2.5, then the UI displays an explicit no-data/unsupported state and no fabricated value or unrelated-location value.*

*AC-F07-04: Given historical/map values exist for a supported location and requested time range, when a Web User opens the map/history view, then displayed values correspond only to that location and time range and retain the source/timestamp provenance of the underlying records.*

*AC-F07-05: Given the dashboard provider API returns HTTP 4xx/5xx or times out, when a Web User refreshes the dashboard, then the UI shows an understandable unavailable/stale state, does not fabricate a new measurement, and exposes a retry action.*

Additional draft criteria — FR-11–FR-13

Proposed for review; priorities and release scope remain unchanged. These criteria are not test results.

AC-F03-01 (DRAFT): Given comparable baseline and ML evaluation results, when the evaluation view is opened, then show each model, metric, unit, dataset/evaluation period and result. The metric and comparison protocol require team confirmation.

AC-F03-02 (DRAFT): Given a model result is missing or evaluation conditions differ, when results are viewed, then identify the missing/non-comparable data without substituting zero or claiming model superiority.

AC-F08-01 (DRAFT): Given a valid AQI under the standard approved in BR-01, when interpretation guidance is viewed, then show the corresponding category and approved general guidance, clearly identified as non-medical.

AC-F08-02 (DRAFT): Given AQI is missing/invalid or the applicable standard is unknown, when guidance is viewed, then state that classification is unavailable and do not present guidance as though air quality were Good.

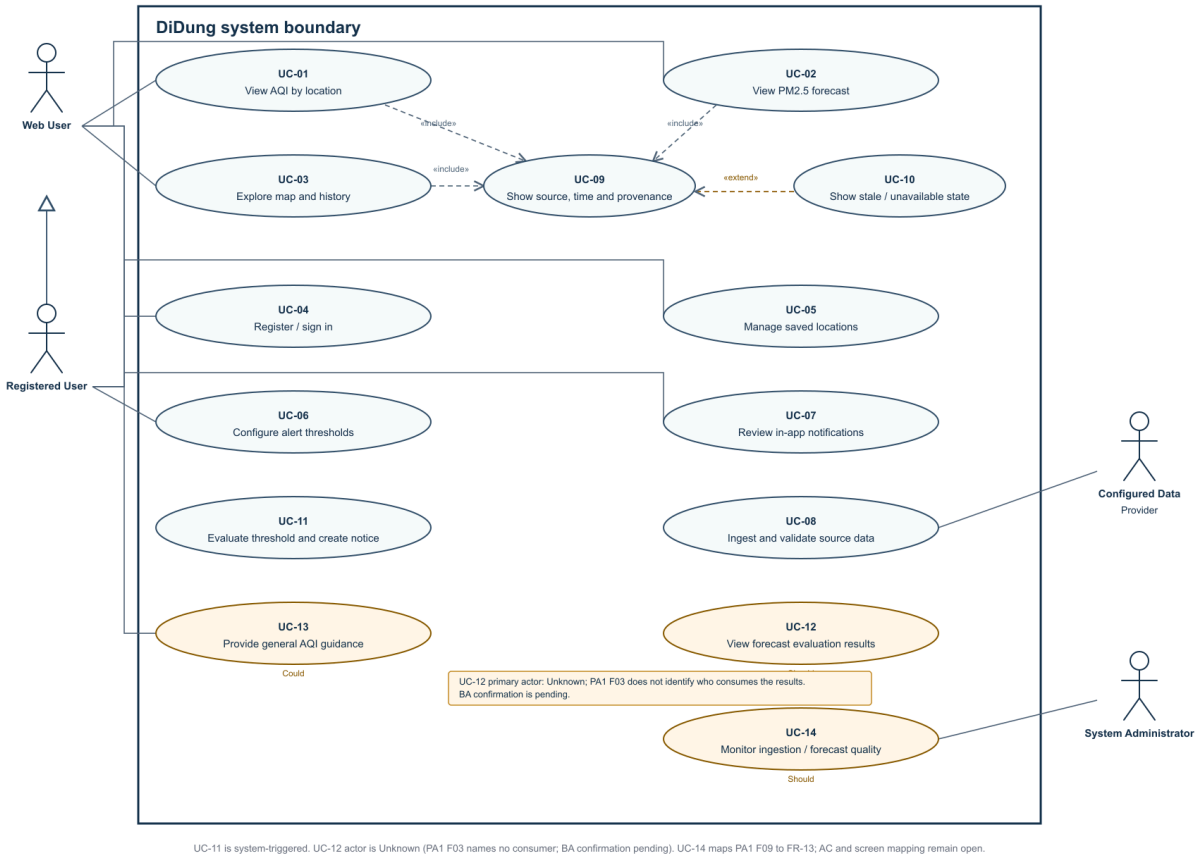
AC-F09-01 (DRAFT): Given an authorized System Administrator and available monitoring data, when the monitoring view is opened, then show pipeline status, stale-data warnings, deployment health and forecast-error summaries with source/time; mark missing data unavailable.

AC-F09-02 (DRAFT): Given an unauthenticated or non-admin requester, when protected monitoring data are requested, then deny access with 401/403 as applicable and disclose no protected monitoring content.

*All AC are draft and subject to BA/Tester review.*

# 6 Use Case Model

## 6.1 Use Case Diagram




UC-11 is system-triggered. UC-12 actor is Unknown (PA1 F03 names no consumer; BA confirmation pending). UC-14 maps PA1 F09 to FR-13; AC and screen mapping remain open.

## 6.2 Actors

| Actor | Description |
|---|---|
| Web User | Unauthenticated visitor who views public AQI/PM2.5, forecasts, map/history and guidance. |
| Registered User | Authenticated Web User (specializes Web User); manages saved locations, thresholds and in-app notices. |

| Actor | Description |
|---|---|
| Configured Data Provider | External system supplying permitted air-quality/weather inputs. The owner reports the production map uses Open-Meteo/CAMS model data; the exact configured model/domain, returned grid, API payload and ingestion behavior remain pending runtime validation. |
| System Administrator | System Administrator reviews pipeline run status, stale-data warnings, deployment health and forecast-error summaries (PA1 F09, Should). |

## 6.3 Use Case List

| ID | Use case | Primary actor | Priority | Related FR |
|---|---|---|---|---|
| UC-01 | View AQI by location | Web User | Must | FR-03, FR-10 |
| UC-02 | View PM2.5 forecast | Web User | Must | FR-05, FR-10 |
| UC-03 | Explore map and history | Web User | Must | FR-04, FR-10 |
| UC-04 | Register / sign in | Web User | Must | FR-06 |
| UC-05 | Manage saved locations | Registered User | Must | FR-07 |
| UC-06 | Configure alert thresholds | Registered User | Must | FR-07 |
| UC-07 | Review in-app notifications | Registered User | Must | FR-09 |
| UC-08 | Ingest and validate source data | None - automatic process; external provider is supporting actor | Must | FR-01, FR-02 |
| UC-09 | Show source, time and provenance | Web User | Must | FR-03, FR-04, FR-05; BR-02 |
| UC-10 | Show stale / unavailable state | Web User | Must | FR-10 |
| UC-11 | Evaluate threshold and create in-app notice | None - automatic evaluation; Registered User is recipient | Must | FR-08 |
| UC-12 | View forecast evaluation results | Unknown | Should | FR-11 |

| ID | Use case | Primary actor | Priority | Related FR |
|---|---|---|---|---|
| UC-13 | Provide general AQI guidance | Web User | Could | FR-12 |
| UC-14 | Monitor ingestion / forecast quality | System Administrator | Should (PA1 F09) | FR-13 |

# 7 Use Case Specifications

*Fourteen use cases are listed in Section 6. Draft specifications for the eleven Must-have use cases, UC-01–UC-11, are provided below. UC-12 and UC-14 are Should-have and UC-13 is Could-have; their detailed specifications remain pending BA/team review. Suggested answers to unresolved modeling questions are explicitly labeled as proposals. BR-01, BR-03 and BR-05 decisions remain pending; no project plan, technical configuration or approval is changed by these drafting updates.*

## 7.1 UC-01 - View AQI by Location

| *Use case ID* | UC-01 |
|---|---|
| *Use case name* | View AQI by location |
| *Brief description* | Web User selects a supported location and sees available AQI/PM2.5 together with provenance and data state. |
| *Primary actor* | Web User |
| *Secondary actors* | Configured data provider (indirectly via the system) |
| *Priority* | Must |
| *Related requirements* | FR-03, FR-10; BR-01 to BR-04 |
| *Trigger* | Web User opens the dashboard or selects a location. |
| *Pre-conditions* | Web app available; supported locations configured. |
| *Post-conditions* | Selected location and available values displayed, or honest unavailable/stale status. |
| *Main flow* | *1. Web User opens AQI dashboard.*<br>*2. System displays supported location options.*<br>*3. Web User selects a location.*<br>*4. System queries the most appropriate available data.*<br>*5. System checks freshness and provenance.*<br>*6. System shows AQI, PM2.5, source, time and status.* |

| *Use case ID* | UC-01 |
|---|---|
| *Alternative flows* | *A1 (step 3): visitor searches for another supported location; continue at step 4.*<br>*A2 (step 5): source is stale; label stale and show original timestamp.* |
| *Exception flows* | *E1 (step 4): provider/storage error; display unavailable state and retry option.*<br>*E2 (step 3): unsupported location; explain supported coverage.* |
| *Business rules* | BR-01, BR-02, BR-03, BR-04 |
| *Non-functional constraints* | NFR-01, NFR-04, NFR-05, NFR-07 |

## 7.2 UC-02 - View PM2.5 Forecast

| *Use case ID* | UC-02 |
|---|---|
| *Use case name* | View PM2.5 Forecast |
| *Brief description* | A Web User views available short-term PM2.5 forecasts for a selected supported location, with the forecast period, provenance and available time metadata, or an explicit unavailable state. |
| *Primary actor* | Web User |
| *Secondary actors* | None during this viewing interaction. |
| *Priority* | Must (draft Scope A) |
| *Related requirements* | FR-05, FR-10 |
| *Trigger* | User opens forecast for a supported location. |
| *Pre-conditions* | A supported location has been selected. |
| *Post-conditions* | Forecast output is shown with issue time/horizon, or an unavailable status. |
| *Main flow* | *1. Open forecast view.*<br>*2. Select location and horizon.*<br>*3. Retrieve available forecast.*<br>*4. Validate model output, issue time and horizon.*<br>*5. Render forecast chart and provenance.* |
| *Alternative flows* | *A1 (at main step 2): Web User chooses another supported forecast horizon.*<br>*A1.1 Repeat retrieval from main step 3 under the existing supported-horizon rule.* |
| *Exception flows* | *E1 (at main step 3): no forecast is available or retrieval fails.*<br>*E1.1 Explain unavailability without inventing a curve.*<br>*E1.2 Return to the forecast view; an applicable retry repeats main step 3.* |
| *Business rules* | BR-02, BR-03, BR-04 |
| *Non-functional constraints* | NFR-01, NFR-05, NFR-07 |

## 7.3 UC-03 - Explore Map and History

| *Use case ID* | UC-03 |
|---|---|
| *Use case name* | Explore Map and History |
| *Brief description* | A Web User explores air-quality information on the map and available historical values for a selected supported location and time range, with source and time labels. |
| *Primary actor* | Web User |
| *Secondary actors* | None during this viewing interaction. |
| *Priority* | Must (draft Scope A) |
| *Related requirements* | FR-04, FR-10 |
| *Trigger* | User opens map or history view. |
| *Pre-conditions* | A supported region is configured. |
| *Post-conditions* | Map/history is associated with selected place/time, or no-data state is shown. |
| *Main flow* | *1. Open map/history.*<br>*2. Select location/time range.*<br>*3. Request available records.*<br>*4. Label time, source and data coverage.*<br>*5. Render records on the map/chart.* |
| *Alternative flows* | *A1 (at main step 2): Web User changes the requested time range or supported region.*<br>*A1.1 Repeat the request from main step 3.* |
| *Exception flows* | *E1 (at main step 3): no historical records are available.*<br>*E1.1 Display an explicit no-data status without fabricating values.*<br>*E1.2 Return to selection at main step 2.*<br>*E2 (at main step 3): retrieval fails.*<br>*E2.1 Show an understandable unavailable/error state.*<br>*E2.2 An applicable retry restarts main step 3.* |
| *Business rules* | BR-02, BR-04 |

| *Use case ID* | UC-03 |
|---|---|
| *Non-functional constraints* | NFR-04, NFR-07 |

## 7.4 UC-04 - Sign in or Register

| *Use case ID* | UC-04 |
|---|---|
| *Use case name* | Sign in or Register |
| *Brief description* | A Web User registers or signs in through the configured identity service to access account-specific features; unsuccessful requests do not disclose protected data. |
| *Primary actor* | Web User |
| *Secondary actors* | Suggested actor: Configured Identity Service (Supabase Auth), if treated as external to DiDung. Pending BA confirmation of the system boundary; this proposal does not change the team's authentication architecture. |
| *Priority* | Must (draft Scope A) |
| *Related requirements* | FR-06 |
| *Trigger* | Visitor wants personal preferences. |
| *Pre-conditions* | Auth service configured; registration enabled if offered. |
| *Post-conditions* | Authenticated session created, or a safe error presented. |
| *Main flow* | *1. Open sign-in screen.*<br>*2. Enter credentials.*<br>*3. Submit form.*<br>*4. System validates input and sends authentication request.*<br>*5. On success, establish authorized session and open user area.* |
| *Alternative flows* | *A1 (at main step 2): Web User chooses registration.*<br>*A1.1 Complete the configured sign-up flow and any verification required by the identity service.*<br>*A1.2 If verification is still pending, show the next required action and end without claiming an authenticated session.*<br>*A1.3 Otherwise continue to main step 5 only after authentication succeeds.* |

| *Use case ID* | UC-04 |
|---|---|
| *Exception flows* | *E1 (at main step 4): details are rejected or authentication is unavailable.*<br>*E1.1 Deny access and show a non-secret error.*<br>*E1.2 Allow another attempt from main step 2.*<br>*E2 (alternative entry trigger): an unauthenticated Web User attempts to open a protected route.*<br>*E2.1 Direct the user to sign in and continue at main step 1; no protected information is disclosed.* |
| *Business rules* | BR-06 |
| *Non-functional constraints* | NFR-02, NFR-08 |

## 7.5 UC-05 - Manage Saved Locations

| *Use case ID* | UC-05 |
|---|---|
| *Use case name* | Manage Saved Locations |
| *Brief description* | A Registered User adds, views, changes or removes supported locations saved under their own account. |
| *Primary actor* | Registered User |
| *Secondary actors* | Suggested answer: None for this authenticated account interaction within the DiDung boundary. Pending BA confirmation. |
| *Priority* | Must (draft Scope A) |
| *Related requirements* | FR-07 |
| *Trigger* | User adds/edits/removes a followed location. |
| *Pre-conditions* | User is authenticated. |
| *Post-conditions* | Preference saved for correct user, or validation error reported. |
| *Main flow* | *1. Open account preferences.*<br>*2. View saved locations.*<br>*3. Choose add/edit/delete.*<br>*4. Validate supported location.*<br>*5. Save changes and confirm updated list.* |

| *Use case ID* | UC-05 |
|---|---|
| *Alternative flows* | *A1 (at main step 3): Registered User edits an existing saved location.*<br>*A1.1 Continue with validation at main step 4 and persist the accepted change at step 5.*<br>*A2 (at main step 3): Registered User deletes an existing saved location.*<br>*A2.1 Persist the authorized deletion and show the updated list at main step 5.* |
| *Exception flows* | *E1 (at main step 4): the location is unsupported.*<br>*E1.1 Show the validation issue and return to main step 3.*<br>*E2 (at main step 5): persistence fails.*<br>*E2.1 Show the failure, preserve the previously saved data and do not claim success.*<br>*E2.2 Return to the saved-location view.* |
| *Business rules* | BR-06 |
| *Non-functional constraints* | NFR-02, NFR-08 |

## 7.6 UC-06 - Configure Alert Threshold

| *Use case ID* | UC-06 |
|---|---|
| *Use case name* | Configure Alert Threshold |
| *Brief description* | A Registered User creates or updates a location-specific alert threshold and manages its active status for subsequent automatic evaluation. |
| *Primary actor* | Registered User |
| *Secondary actors* | Suggested answer: None for this authenticated account interaction within the DiDung boundary. Pending BA confirmation. |
| *Priority* | Must (draft Scope A) |
| *Related requirements* | FR-07 |
| *Trigger* | User configures a personal notification threshold. |
| *Pre-conditions* | User is signed in; alert metrics and comparator rules defined. |

| *Use case ID* | UC-06 |
|---|---|
| *Post-conditions* | Threshold is stored, used by eligible evaluations, or rejected as invalid. |
| *Main flow* | *1. Open alert settings.*<br>*2. Choose a tracked location and supported metric.*<br>*3. Enter a valid threshold.*<br>*4. Validate domain and comparator.*<br>*5. Save threshold to the user's account.*<br>*6. Subsequent eligible data may be evaluated against it.* |
| *Alternative flows* | *A1 (at main step 3): Registered User modifies an existing threshold.*<br>*A1.1 Validate the proposed value at main step 4 and save it at step 5.*<br>*A2 (at main step 3): Registered User disables an existing rule.*<br>*A2.1 Save the active-status change and confirm it only if persistence succeeds.*<br>*A2.2 Return to alert settings.* |
| *Exception flows* | *E1 (at main step 4): the threshold is invalid under the approved rule.*<br>*E1.1 Show an actionable validation message and return to main step 3.*<br>*E2 (at main step 5): storage fails.*<br>*E2.1 Show the failure without claiming the settings were saved.*<br>*E2.2 Return to alert settings.* |
| *Business rules* | BR-05, BR-06 |
| *Non-functional constraints* | NFR-02, NFR-08 |

## 7.7 UC-07 - Review In-App Notifications

| *Use case ID* | UC-07 |
|---|---|
| *Use case name* | Review In-App Notifications |
| *Brief description* | A Registered User views their own in-app notifications and marks individual notices as read, with persisted read status and account isolation. |
| *Primary actor* | Registered User |

| *Use case ID* | UC-07 |
|---|---|
| *Secondary actors* | Suggested answer: None for this authenticated account interaction within the DiDung boundary. Pending BA confirmation. |
| *Priority* | Must (draft Scope A) |
| *Related requirements* | FR-09 |
| *Trigger* | User wishes to inspect generated alerts. |
| *Pre-conditions* | User is signed in; notice history may be empty. |
| *Post-conditions* | Authorized notice list shown; read state may be updated. |
| *Main flow* | *1. Open notification center.*<br>*2. Retrieve current user's notices.*<br>*3. Display event metric, time and location.*<br>*4. User opens a notice.*<br>*5. User marks it as read.*<br>*6. System saves read state.* |
| *Alternative flows* | *A1 (at main step 2): there are no notices for the account.*<br>*A1.1 Show an empty-state message.*<br>*A1.2 The use case ends without modifying stored notices.* |
| *Exception flows* | *E1 (at main step 2): the request is unauthorized.*<br>*E1.1 Deny access without exposing notification content.*<br>*E1.2 The use case ends.*<br>*E2 (at main step 6): the update fails.*<br>*E2.1 Show the failure without falsely marking the notice read.*<br>*E2.2 Return to the notice view; retry repeats the read-state update.* |
| *Business rules* | BR-05, BR-06 |
| *Non-functional constraints* | NFR-02, NFR-08 |

## 7.8 UC-08 - Ingest and Validate Source Data

| *Use case ID* | UC-08 |
|---|---|
| *Use case name* | Ingest and Validate Source Data |

| *Use case ID* | UC-08 |
|---|---|
| *Brief description* | The system retrieves permitted pollutant/weather inputs on the configured schedule, validates the required data and stores accepted records with provenance while recording failures. |
| *Primary actor* | None - internal scheduled process |
| *Secondary actors* | Configured data provider |
| *Priority* | Must (draft Scope A) |
| *Related requirements* | FR-01, FR-02 |
| *Trigger* | Scheduled ingestion trigger fires. |
| *Pre-conditions* | Source adapter and authorized API configuration exist. |
| *Post-conditions* | Valid records are stored with provenance; malformed inputs flagged. |
| *Main flow* | *1. Scheduler initiates job.*<br>*2. Adapter requests permitted provider data.*<br>*3. System parses location, metric, units and time.*<br>*4. Validate numeric values, freshness and schema.*<br>*5. Save valid records with provider metadata.*<br>*6. Log outcome and rejected records for inspection.* |
| *Alternative flows* | *A1 (at main step 2): the configured provider requires paging under the existing source policy.*<br>*A1.1 Process permitted pages within the configured limits, then continue at main step 3.*<br>*Draft clarification: this conditional source behavior remains subject to source validation; it does not prescribe a new retrieval design.* |
| *Exception flows* | *E1 (at main step 2): the provider times out or returns a quota error.*<br>*E1.1 Record the failure and apply only the existing configured retry policy.*<br>*E1.2 Do not create a fabricated valid measurement; end the unsuccessful attempt.*<br>*E2 (at main step 4): a record is invalid.*<br>*E2.1 Flag/reject it under the configured validation rule; do not store it as valid or fabricate zero values.*<br>*E2.2 Continue processing remaining records, then record the outcome at main step 6.* |

| *Use case ID* | UC-08 |
|---|---|
| *Business rules* | BR-02, BR-03, BR-04 |
| *Non-functional constraints* | NFR-05, NFR-06 |

## 7.9 UC-09 - Show Source, Time and Provenance

| Use case ID | UC-09 |
|---|---|
| Use case name | Show Source, Time and Provenance |
| Brief description | A Web User views source, time and provenance labels associated with the air-quality values or forecasts displayed in UC-01, UC-02 and UC-03. |
| Primary actor | Web User |
| Secondary actors | Proposed: None during this display interaction; the configured provider remains associated with UC-08. Actor-boundary interpretation pending BA confirmation. |
| Priority | Must (draft Scope A) |
| Related requirements | FR-03, FR-04, FR-05; BR-02 |
| Trigger | A parent interaction prepares an available value or series for display. |
| Pre-conditions | A supported location and viewing context are selected; the parent interaction has retrieved the available data and metadata. |
| Post-conditions | Available values carry truthful source/time/provenance labels; missing metadata are disclosed rather than invented. |
| Main flow | 1. System receives the available record or series and metadata from the parent interaction.<br>2. System identifies the location, metric, unit, source and observed/modeled status.<br>3. System identifies the relevant data time, and the forecast period/issue time where available and required.<br>4. System formats the available times with an explicit time zone.<br>5. System displays the labels with the corresponding values or chart.<br>6. Control returns to the parent interaction. |

| Use case ID | UC-09 |
|---|---|
| Alternative flows | A1 (at main step 2): data are modeled.<br>A1.1 Label the data as modeled rather than as a station measurement.<br>A1.2 Resume at main step 3.<br>A2 (at main step 3): a required metadata field is missing.<br>A2.1 Show that the metadata are unavailable; do not invent a source or time.<br>A2.2 Proposed handling: if the approved rule requires that field before displaying a value, show the unavailable state through UC-10; otherwise display the limitation and resume at main step 4. This choice remains pending BA confirmation. |
| Exception flows | E1 (at main step 1): no usable record is available or retrieval failed.<br>E1.1 Display the appropriate state through UC-10 without fabricating a value.<br>E1.2 Return control to the parent interaction. |
| Business rules | BR-02, BR-03, BR-04. Pending clarification: which metadata fields are mandatory for each view. Data valid time, fetch time and model issue time must not be treated as interchangeable. |
| Non-functional constraints | NFR-04, NFR-06, NFR-07 |

## 7.10 UC-10 - Show Stale / Unavailable State

| Use case ID | UC-10 |
|---|---|
| Use case name | Show Stale / Unavailable State |
| Brief description | A Web User receives an explicit stale, no-data or unavailable/error state when requested air-quality information cannot be shown as current and usable. |
| Primary actor | Web User |
| Secondary actors | Proposed: None for this display behavior; pending BA confirmation of the system boundary. |
| Priority | Must (draft Scope A) |

| Use case ID | UC-10 |
|---|---|
| Related requirements | FR-10; invoked from the relevant public data-view interactions. |
| Trigger | A parent interaction detects stale data, no usable data, an unsupported location or a retrieval failure. |
| Pre-conditions | A request context and its data/result status are available for classification. |
| Post-conditions | The user can distinguish stale data from missing data and request errors; no fabricated current value or curve is displayed. |
| Main flow | 1. System receives the data/result status from the parent interaction.<br>2. System classifies the condition under the approved freshness and availability rules.<br>3. Where an available record is classified as stale, system labels it stale and retains its true timestamp and provenance.<br>4. System offers the applicable refresh or location-selection action.<br>5. Control returns to the parent interaction with the data state visible. |
| Alternative flows | A1 (at main step 2): no usable records exist.<br>A1.1 Show an explicit no-data state without inventing a value or curve.<br>A1.2 Return to the parent interaction.<br>A2 (at main step 2): the selected location is known to be unsupported.<br>A2.1 Explain the supported coverage.<br>A2.2 Return to location selection.<br>A3 (at main step 4): the user retries.<br>A3.1 Restart retrieval in the parent interaction.<br>A3.2 Classify the new result again where necessary. |
| Exception flows | E1 (at main step 2): the data request fails or times out.<br>E1.1 Show an understandable unavailable/error state and the applicable retry action.<br>E1.2 Do not fabricate a replacement value; return to the parent interaction.<br>E2 (at main step 2): required time metadata are missing or invalid.<br>E2.1 Do not assert that the value is fresh.<br>E2.2 Proposed handling: show a metadata limitation or unavailable state according to the approved rule, then return control. Final choice pending BA confirmation. |

| Use case ID | UC-10 |
|---|---|
| Business rules | BR-02, BR-03, BR-04. Freshness thresholds and source-specific treatment remain pending under BR-03; this specification does not select a numeric threshold or change forecast/data policy. |
| Non-functional constraints | NFR-04, NFR-05, NFR-07 |

## 7.11 UC-11 - Evaluate Threshold and Create In-App Notice

| Use case ID | UC-11 |
|---|---|
| Use case name | Evaluate Threshold and Create In-App Notice |
| Brief description | The system evaluates eligible air-quality data against active user thresholds and creates an account-specific in-app notice when the configured and approved trigger conditions are satisfied. |
| Primary actor | None - internal automatic evaluation; Registered User is the recipient. |
| Secondary actors | Proposed: None during evaluation of stored data; pending BA confirmation of the system boundary. |
| Priority | Must (draft Scope A) |
| Related requirements | FR-08. Threshold configuration is described in UC-06; input validation is described in UC-08. |
| Trigger | The configured automatic evaluation trigger fires. |
| Pre-conditions | An active threshold identifies its owner, supported location and metric. Comparator, input eligibility and duplicate-suppression rules must be confirmed under BR-03/BR-05 before acceptance. |
| Post-conditions | Qualifying eligible events create notices only for the correct account and subject to the approved duplicate rule. Non-qualifying/ineligible inputs create no notice; failed operations are not reported as successful. |
| Main flow | 1. System starts an authorized evaluation run through the configured trigger. |

| Use case ID | UC-11 |
|---|---|
| | 2. System loads active thresholds with their owning accounts, locations and metrics.<br>3. System obtains eligible data for each threshold's location and metric.<br>4. System evaluates the value against the approved trigger rule.<br>5. For a qualifying result, system checks the configured duplicate-suppression identity and scope.<br>6. If not suppressed, system stores an unread in-app notice for the threshold owner, with the required location, value and trigger-time information.<br>7. System records the run outcome and ends. |
| Alternative flows | A1 (at main step 2): no active thresholds exist.<br>A1.1 Complete the run without creating notices.<br>A1.2 The use case ends.<br>A2 (at main step 3): the input is missing, stale or invalid under the approved eligibility rule.<br>A2.1 Create no notice from that input.<br>A2.2 Continue with the next threshold, or proceed to main step 7 when none remain.<br>A3 (at main step 4): the trigger condition is not satisfied.<br>A3.1 Create no notice for that evaluation.<br>A3.2 Continue with the next threshold, or proceed to main step 7.<br>A4 (at main step 5): the event is suppressed as a duplicate.<br>A4.1 Retain the existing notice without creating an additional notice.<br>A4.2 Continue with the next threshold, or proceed to main step 7. |
| Exception flows | E1 (at main step 1): the trigger request is unauthorized.<br>E1.1 Reject the request and create no notices from it.<br>E1.2 The use case ends.<br>E2 (at main step 2, 3 or 6): a required read/write operation fails.<br>E2.1 Record the failure without claiming that the failed operation succeeded.<br>E2.2 End the run under the existing error-handling policy; no rollback or retry mechanism is prescribed here. |
| Business rules | BR-03, BR-05, BR-06. Pending decisions remain unchanged: comparator/equality boundary, crossing versus threshold satisfaction, eligible-input rules, event identity and suppression window. Suggested drafting approach: document the team's confirmed rule in BR-05 and reference it here; do not infer approval from the implementation. |

| Use case ID | UC-11 |
|---|---|
| Non-functional constraints | NFR-02, NFR-05, NFR-08 |

# 8 Prototype / Mockups

*Design reference (Designer DID-50/51/52):* *Figma*

| Screen | Related use case(s) | Mockup / link |
|---|---|---|
| SCR-01 Landing | UC-01, UC-02 | DiDung design |
| SCR-02 AQI Dashboard | UC-01, UC-09, UC-10 | Figma |
| SCR-03 Forecast | UC-02, UC-09, UC-10 | Figma |
| SCR-04 Map & History | UC-03, UC-09, UC-10 | Figma |
| SCR-05 Sign in | UC-04 | Figma |
| SCR-06 Alert settings | UC-05, UC-06 | Figma |
| SCR-07 Notification center | UC-07 | Figma |

# 9 Requirements Traceability Matrix

| Req. ID | Source / Priority | Use case | Acceptance Criteria | Screen | Design element (PA3) | Test cases (PA4) |
|---|---|---|---|---|---|---|
| FR-01 | PA1 F01<br>Must<br>DID-43 | UC-08 | AC-F01-01<br>AC-F01-03<br>AC-F01-05 | Background ingestion; no direct user screen. | TBD - PA3 | TBD - PA4 |
| FR-02 | PA1 F01<br>Must<br>BR-03 pending | UC-08<br>UC-10<br>UC-11 | AC-F01-01<br>AC-F01-02<br>AC-F01-04<br>AC-F01-05<br>AC-F06-06 | Background validation; SCR-02–04 for status.<br>State reference | TBD - PA3 | TBD - PA4 |
| FR-03 | PA1 F07<br>Must | UC-01<br>UC-09<br>UC-10 | AC-F07-01<br>AC-F07-02<br>AC-F07-03<br>AC-F07-05 | SCR-02 AQI dashboard<br>Desktop \| Mobile | TBD - PA3 | TBD - PA4 |
| FR-04 | PA1 F07<br>Must | UC-03<br>UC-09<br>UC-10 | AC-F07-04<br>Error AC: pending | SCR-04 Map & history<br>Desktop \| Mobile | TBD - PA3 | TBD - PA4 |
| FR-05 | PA1 F02<br>Must | UC-02<br>UC-09<br>UC-10 | AC-F02-01<br>AC-F02-02 | SCR-03 Forecast<br>Desktop \| Mobile<br>Coverage: see notes. | TBD - PA3 | TBD - PA4 |
| FR-06 | PA1 F05<br>Must | UC-04 | AC-F05-01<br>AC-F05-02<br>AC-F05-04<br>AC-F05-05 | SCR-05 Sign in<br>Desktop \| Mobile<br>SCR-06: protected view. | TBD - PA3 | TBD - PA4 |
| FR-07 | PA1 F05<br>Must | UC-05<br>UC-06 | AC-F05-03<br>AC-F05-05<br>AC-F05-06 | SCR-06 Thresholds<br>Desktop \| Mobile<br>Saved locations: pending. | TBD - PA3 | TBD - PA4 |
| FR-08 | PA1 F06<br>Must | UC-11<br>UC-06 (support) | AC-F06-01<br>AC-F06-02<br>AC-F06-05<br>AC-F06-06 | Background evaluator.<br>SCR-06: configuration;<br>SCR-07: output. | TBD - PA3 | TBD - PA4 |
| FR-09 | PA1 F06<br>Must | UC-07 | AC-F06-03<br>AC-F06-04<br>AC-F06-07 | SCR-07 Notifications<br>Desktop \| Mobile | TBD - PA3 | TBD - PA4 |

| Req. ID | Source / Priority | Use case | Acceptance Criteria | Screen | Design element (PA3) | Test cases (PA4) |
|---|---|---|---|---|---|---|
| FR-10 | PA1 F07 Must | UC-01–03 UC-10 | AC-F02-02 AC-F07-02 AC-F07-03 AC-F07-05 | SCR-02–04 variants State reference | TBD - PA3 | TBD - PA4 |
| FR-11 | PA1 F03 Should | UC-12 | DRAFT: AC-F03-01 AC-F03-02 | Pending: baseline/ML evaluation screen not identified. | TBD - PA3 | TBD - PA4 |
| FR-12 | PA1 F08 Could | UC-13 | DRAFT: AC-F08-01 AC-F08-02 | Proposed: SCR-02/03 guidance sections. Coverage: see notes. | TBD - PA3 | TBD - PA4 |
| FR-13 | PA1 F09 Should | UC-14 | DRAFT: AC-F09-01 AC-F09-02 | Pending: admin monitoring screen not identified. | TBD - PA3 | TBD - PA4 |

**Coverage notes — draft**

FR-05: SCR-03 and the current chart use 24-hour AQI, while AC-F02-01 requires a PM2.5 forecast at 1–6 hours with issue time. Figma issue times are illustrative. Full conformance remains pending.

FR-07: SCR-06 covers thresholds; a location selector or localStorage setting does not establish account-owned saved-location CRUD. Its screen/API evidence remains pending.

FR-08: Figma uses >; current code uses ≥, a 3-hour input window and per-threshold/day UTC deduplication. Crossing is not established. BR-03/05 must resolve freshness, equality, trigger semantics and event identity.

FR-12: SCR-02/03 contain candidate interpretation guidance; coverage depends on the AQI standard approved under BR-01. FR-11/13 screens and the optional delivery/deferment decision remain pending.

Traceability: FR-04 covers map/history; FR-05 covers forecast. UC-10 links represent unavailable/stale/error branches. All six additional AC remain DRAFT. PA3/PA4 remain TBD.

Evidence: Figma frame links above; source snapshot d73a715; DID-43 (source assessment), DID-50 (inventory), DID-51 (prototype), DID-52 (accessibility). Source/design inspection is not runtime acceptance. Independent approval of this RTM revision remains pending.

# 10 Requirements Review

*Requirements Review status: IN PROGRESS while BA feedback and remaining information from Nguyễn Thành Đạt are pending. The AI-assisted desk review of this draft is complete; the project owner is reviewing the updated draft. Hiếu and the third reviewer are recorded as having completed their review and Approved under the project-owner rule because no high-impact issue was identified. The third reviewer's identity is Unknown. TA is not a project participant and no TA approval is required. No meeting is claimed.*

| # | Issue found | Requirement(s) | Resolution |
|---|---|---|---|
| 1 | Scope A differs from flood/tide Must-haves in PA0 | BR-07, scope | Resolved for the PA2 draft: follow PA1 where it supersedes PA0, per project-owner direction. TA is not a project participant and no TA approval is required. |
| 2 | AQI scale and PM2.5-to-AQI conversion source undecided | BR-01, FR-03 | Pending BA response on the AQI scale/calculation source. The current map displays US AQI; the adopted standard and calculation source remain unconfirmed until BA feedback. |
| 3 | Open-Meteo production map reports AQI/PM2.5 at each location; nearby pins may share modeled values | FR-01, BR-03 | Partially assessed: owner confirms every app map location has AQI/PM2.5. Screenshot shows one popup (US AQI 84, moderate, PM2.5 33.3 µg/m³, 23:00 01/11/2026 UTC+7); read-only review of deployed interface confirms shown location entries list AQI/category, PM2.5 and effective time, with many shared values. The map labels Open-Meteo/CAMS model data and says nearby points may share an index. This confirms display behavior, not API payloads, ingestion or distinct sensor readings. Official docs: coordinate-based PM2.5 in µg/m³; global CAMS ~45 km, underlying 3-hourly, updated every 12 hours; API hourly series and up to 7 forecast days. `past_days` accepts 0–92, but completeness or observed-history semantics are not established. Hiếu's source assessment is recorded as reviewed and Approved; no high-impact mismatch was identified for the explicitly labeled model-based location display. Remaining project evidence is pending: verify the deployed model/domain and |

| # | Issue found | Requirement(s) | Resolution |
|---|---|---|---|
| | | | HCMC responses, returned grid coordinates, FR-04 history suitability, quota, non-commercial licence fit and Open-Meteo/CAMS attribution. Free terms: <10,000 calls/day, 5,000/hour, 600/minute; Vercel may hit limits earlier; CC BY 4.0. Sources: https://open-meteo.com/en/docs/air-quality-api ; https://open-meteo.com/en/docs/model-updates ; https://open-meteo.com/en/terms. |
| 4 | NFR numeric targets need feasibility validation | NFR-01-08 | Open - verify targets with actual environment and tester. |
| 5 | Prototype, screen IDs and accessibility audit not confirmed | FR-03-10; Section 8 | Open - Designer DID-50-52; Figma and review links pending. |
| 6 | Use Case Diagram and UC-01-UC-14 require UML review | UC-01-UC-14 | Current draft: specifications for all eleven Must-have use cases, UC-01–UC-11, are present. UC-12–UC-14 detailed specifications remain pending the optional-scope decision. Review of the current diagram, actor boundaries, relationships and revised specifications remains Pending. Earlier owner-reported approvals apply to prior revisions and are not extended to these additions. Draw.io/ZIP access remains recorded as read-only; no new meeting or approval is claimed. BR-01/03/05 decisions remain pending. |
| 7 | Current SRS review: Hiếu/third reviewer complete and Approved; BA response Pending | All SRS sections | The project owner earlier reported BA approval of a previous PA2 baseline; BA feedback on this SRS revision is Pending, with no BA-authored reply on the current version. Hiếu and the third reviewer are recorded as having completed review and Approved under the owner rule; no high-impact issue was identified. DID-54 was checked In review on 2026-11-02; that workflow status is not approval. No meeting participants/date are recorded, and no meeting is claimed. |
| 8 | Contribution allocation confirmed; evidence compilation pending | Section 1 | Allocation confirmed on 02 Nov 2026: Hiếu 25%, Tiến 25%, Đạt 25%, Hào 25%; total |

| # | Issue found | Requirement(s) | Resolution |
|---|---|---|---|
| | | | 100%. Section 1 has been updated. Final artifact/review links per member remain to be compiled; the percentages do not certify completion of every assigned task. |
| 9 | RTM acceptance-criteria and screen gaps for FR-11–FR-13 | FR-11–FR-13; UC-12–UC-14 | Draft update — 02 Nov 2026: Section 9 now contains frame-level links, coverage gaps and six proposed AC for FR-11–FR-13, defined in Acceptance Criteria. FR-12 has candidate guidance in SCR-02/03; FR-11/13 screen mappings and optional delivery scope remain pending. BA/Tester review and Designer validation of the revised mappings remain pending. Earlier owner-reported approvals apply to the prior revision and are not extended to these edits. PA3/PA4 remain TBD. |
| 10 | No PA2 SRS/RTM PR found in repository audit | DID-57/58; SRS/RTM delivery | Repository audit checked 2026-10-31: the PR list showed 0 open and 10 closed PRs; the main `pa/PA2` directory contained only `.gitkeep`. No matching PA2 SRS/RTM PR, Jira-linked PR, or independent review feedback was found. DID-57 and DID-58 were In Progress; no Moodle receipt was observed. Matching PA2 PR, independent feedback, and Moodle confirmation remain Pending. |
| 11 | 25 Must-have AC drafted; Hiếu/third reviewer approved; BA feedback pending | FR-01, FR-02, FR-05, FR-06, FR-07, FR-09 | 25 Given/When/Then AC cover Must FR-01–FR-10, including source API failure, stale/invalid alert input, unauthorized access, threshold crossing/non-crossing and duplicate suppression. BA feedback on these current scenarios remains Pending; the owner is reviewing the updated draft. Hiếu and the third reviewer are recorded as having completed review and Approved under the owner rule; no high-impact issue was identified. New scenarios remain draft; BR-01/03/05 decisions are pending. |
| 12 | FR-06 crosswalk updated; BA feedback pending; Designer validation pending | FR-06; UC-04; SCR-05/SCR-06 | User approved on 2026-10-31: map FR-06 to UC-04, SCR-05 Sign in and SCR-06 Alert settings. UC-04 exception E2 redirects unauthorized protected-route |

| # | Issue found | Requirement(s) | Resolution |
|---|---|---|---|
| | | | requests to sign in; FR-06 also covers access to account settings. UC-05/UC-06 remain linked to FR-07 only because they specify authenticated preference changes. BA feedback on this current crosswalk is Pending; no BA-authored reply is in Docs comments. Designer validation of the screen mapping remains Pending from Nguyễn Thành Đạt. |

# 11 AI Usage Log

*Entries 1–2 preserve the original AI usage logged by Nguyễn Thành Đạt. Entry 3 records ChatGPT-assisted drafting and revisions directed, reviewed and edited by project owner Võ Đình Cao Minh Hào. Other members must confirm their actual AI usage or explicitly report no use. This log records assisted drafting and owner review; it is not independent verification or direct Hiếu/third-reviewer sign-off.*

| # | Member | Tool | Task / purpose | How the output was verified and changed |
|---|---|---|---|---|
| 1 | Nguyễn Thành Đạt | *ChatGPT* | *Draft PA2 Scope A requirements, use cases, RTM and SRS working copy.* | *Draft cross-checked against PA2 template, PA1 and team choice; independent team validation pending.* |
| 2 | Nguyễn Thành Đạt | ChatGPT | Explain FR/NFR, MoSCoW, AC and Scope Analysis; suggest testable examples. | Examples labeled proposed; user confirms scope A. Numeric NFR limits are not measured. |
| 3 | Võ Đình Cao Minh Hào | ChatGPT | Draft Must-have Given/When/Then acceptance scenarios and review consistency across FRs, use cases, SRS and RTM. | ChatGPT drafted these scenarios; the project owner is the reviewer/editor and will fully review and edit before finalization. Scenarios remain drafts; BA decisions are pending specifically for BR-01/03/05. Hiếu and the third reviewer are recorded as Reviewed/Approved under the owner's rule because no high-impact issue was identified; the third reviewer's identity is Unknown. |
| 4 | Võ Đình Cao Minh Hào | ChatGPT | Draft/revise PA2 use-case diagram and UC-01–UC-14; write UC-09–UC-14 specifications; add AC for FR-11–FR-13; update RTM/review log; inspect Đạt's Figma-linked GitHub documentation. | AI draft checked against PA1 then PA0, current PA2 SRS/RTM and PR #10 by Đạt (entidy06). Editable Draw.io, PlantUML and Mermaid sources plus SVG/PNG exports are synchronized. The acceptance set has 33 draft ACs: 25 Must and 8 for FR-11–FR-13. PR #10 provides a Figma link and frontend spec; actual frame content, accessibility and screen mapping remain unverified. Project owner is the |

| # | Member | Tool | Task / purpose | How the output was verified and changed |
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| | | | | reviewer/editor; this remains a draft for full owner review and editing. BA decisions are pending specifically for BR-01/03/05; UC-12 actor, FR-11 metrics and FR-13 screen/telemetry remain Unknown where sources are silent. Hiếu and the third reviewer are recorded as Reviewed/Approved under the owner's instruction; the third identity is Unknown. No implementation or test pass is claimed. |